

Radio Frequency  
It is a form of air

Types of medium:-

Wireless:-

- \* Radio Frequency

It is a type of wireless medium which uses the Radio signals

Wired:-

- \* Electrical Signals:-

- \* Uses electrical signals as a medium through copper cable

- \* UTP and STP

- \* Light Signals

- \* works through the optic fiber cable

Switch 8-Pins

1 + R

2 - R

3 + T

4 + V

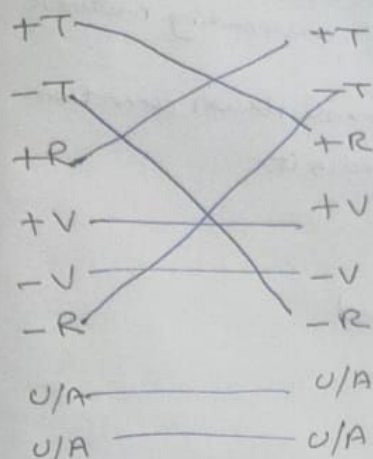
5 - V

6 - T

7 Unassigned Ports

8 Unassigned port

UTP cable Connection between Router and PC?  
Cross cable connection:



Explain Auto MDIX:-

- \* Automatic Medium Dependent Interface Crossover is a feature that is present in the NIC Ports where in the cable connection the <sup>Network Ports Interface</sup> Transmitter and Receiver Pins automatically changes its Role based on the wire connected method.
- \* It Automatically adjust the Tx and Rx Pins and determines whether a straight cable or cross cable is needed.

POE Injector:-

- \* It is a external device that Powers up the Network device via the ethernet connected in between a Connection.
- + So when the cable is passed via the Injector it Powers up the cable where the power is Passed through +V, -V and data through Tx, Rx Separately in a same cable.



## POE Switch:-

- \* It is device which transmits both power and data from the POE Switch to the connected corresponding Network devices.
- \* It does not require any external power connection to power up the devices connected in it.

## <sup>Types</sup> NIC and functions:-

- \* There are two types of NIC card

wired and wireless

### \* wired

- \* There are two types in wired NIC

- \* Electrical NIC Port (ethernet port)

- \* Electrical signal converted to binary and vice versa

- \* Optic Fiber Port

- \* Light signals to binary signals and vice versa

### \* wireless:-

- \* Wifi Adaptors in NIC

- \* Here Radio frequency is converted into Binary and vice versa.